

PATARLAGELE, Romania

WMO: 153280

Lat: 45.3247N

Lon: 26.3694E

Elev: 291

StdP: 97.88

Time Zone: 2.00 (EUE)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.8	-8.4	-14.1	1.1	-7.4	-12.1	1.4	-5.6	12.5	1.5	10.5	0.8	1.3	350	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	31.6	21.3	30.2	20.9	28.8	20.5	23.3	29.3	22.2	28.1	21.3	27.2	1.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.5	16.7	26.5	20.2	15.4	25.3	19.2	14.5	24.5	70.7	28.8	66.6	28.2	63.2	27.4	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.9	6.5	5.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	-0.5	2.3	5.6	10.2	15.8	20.0	21.1	21.3	16.6	10.6	5.6	1.8
	DBStd	8.54	4.31	4.19	4.61	4.12	3.16	2.80	2.86	3.00	3.92	4.03	4.26	4.14
	HDD10.0	1184	327	217	148	48	2	0	0	0	2	42	144	255
	HDD18.3	3011	585	450	395	246	89	14	9	9	79	241	382	512
	CDD10.0	1516	0	1	12	53	183	301	344	352	199	59	12	1
	CDD18.3	301	0	0	0	1	11	65	94	103	26	1	0	0
	CDH23.3	2813	0	0	1	32	143	596	834	918	264	24	1	0
	CDH26.7	873	0	0	0	3	21	159	277	338	74	1	0	0
Wind	WSAvg	1.8	2.1	2.0	2.1	2.0	1.7	1.8	1.6	1.7	1.7	1.5	1.6	1.9
Precipitation	PrecAvg	671	31	25	32	49	79	101	93	76	56	50	39	39
	PrecMax	937	87	75	65	117	169	164	234	170	115	157	96	102
	PrecMin	440	6	2	3	20	26	37	31	17	9	2	3	4
	PrecStd	132	19	17	14	26	38	31	48	41	28	33	23	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.3	18.2	22.2	26.6	29.0	31.7	33.4	33.2	31.3	26.2	20.8	16.1
		MCWB	8.1	11.3	13.2	15.4	19.0	22.7	23.4	21.3	19.7	16.2	13.5	9.8
	2%	DB	10.5	14.6	19.0	23.6	26.3	30.1	31.2	31.7	28.5	23.4	17.1	12.2
		MCWB	6.3	8.7	11.2	14.8	17.9	21.4	22.2	20.6	18.5	15.0	11.4	7.7
	5%	DB	8.3	11.7	16.5	21.0	24.8	28.6	29.7	30.2	26.1	20.9	14.2	10.1
		MCWB	5.0	7.3	9.9	13.1	17.1	20.9	21.7	20.3	17.7	14.0	10.3	7.1
	10%	DB	5.8	9.1	13.6	18.4	23.0	26.9	28.1	28.5	23.7	17.9	11.9	7.9
		MCWB	3.4	5.7	8.6	12.0	16.1	20.1	21.0	19.9	17.0	12.3	9.1	5.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.8	12.2	14.7	16.6	21.9	24.2	25.0	23.6	21.2	17.0	13.9	10.5
		MCDB	13.0	17.2	21.0	24.2	27.6	30.2	30.0	31.6	28.6	24.0	20.0	14.3
	2%	WB	7.0	9.5	12.0	15.4	19.2	22.7	23.6	22.4	19.7	15.6	12.5	8.7
		MCDB	10.0	14.3	18.4	22.9	23.8	28.8	29.1	29.3	26.7	22.4	16.3	11.9
	5%	WB	5.2	7.6	10.3	14.2	18.0	21.5	22.5	21.5	18.6	14.3	11.0	7.0
		MCDB	7.5	11.1	15.5	20.4	23.1	27.6	27.7	28.0	24.8	20.2	13.5	10.1
	10%	WB	3.9	6.0	8.8	12.8	16.9	20.5	21.5	20.6	17.5	12.8	9.5	5.6
		MCDB	5.6	8.7	13.3	17.9	22.0	26.3	26.8	26.8	23.2	17.2	11.7	8.1
Mean Daily Temperature Range	5% DB	MDBR	8.3	10.1	10.8	11.4	11.6	11.5	11.7	12.6	11.7	11.6	8.9	7.8
		MCDBR	12.1	14.9	16.4	16.9	14.6	13.8	13.7	14.8	14.9	16.6	13.3	11.8
		MCWBR	8.5	9.5	9.1	8.5	7.5	6.8	6.5	6.3	6.8	9.0	8.0	7.9
	5% WB	MCDBR	11.7	14.6	15.9	15.9	13.6	13.0	12.1	13.3	13.9	16.3	12.3	11.7
		MCWBR	8.2	9.4	9.0	8.0	7.7	6.9	6.5	6.3	6.7	8.9	7.7	7.6
Clear-Sky Solar Irradiance	taub	0.320	0.350	0.389	0.450	0.426	0.429	0.443	0.458	0.415	0.377	0.348	0.321	
	taud	2.340	2.253	2.212	2.069	2.215	2.242	2.211	2.161	2.247	2.337	2.368	2.359	
	Ebn at Noon	777	816	833	809	845	842	824	794	797	777	735	732	
	Edh at Noon	78	103	123	155	138	135	138	140	118	94	76	70	
All-Sky Solar Radiation	RadAvg	1.32	2.14	3.29	4.48	5.61	5.98	6.09	5.49	3.99	2.62	1.44	1.10	
	RadStd	0.16	0.27	0.35	0.47	0.46	0.45	0.45	0.37	0.35	0.30	0.15	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (68 neighbors)	+0.56	N/A	N/A	+0.66	N/A	N/A	-118	-161	+76	+29

Nomenclature: See separate page